

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1**(Adv-2024-P1-Model) **Date: 04-01-2026**

Time: 09.00Am to 12.00Pm **GTA-10** **Max. Marks: 180**

04-01-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2024-P1)_GTA-10_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | |

For More Join: @IITJEE_Advanced

JEE-ADVANCED-2024-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Matching Type	+3	-1	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Matching Type	+3	-1	4	12
				17	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Matching Type	+3	-1	4	12
Total				**17**	**60**

Sri Chaitanya Educational Institutions
Infinity Learn BY SRI CHAITANYA
40 YEARS


For More Join: @IITJEE_Advanced

MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. Let A, B, C be angles of a triangle with

$$\cos^2 A+\cos^2 B+2\sin A\sin B\cos C=\frac{15}{8} \text{ and } \cos^2 B+\cos^2 C+2\sin B\sin C\cos A=\frac{14}{9}$$

which of the following is CORRECT?

A) $\sin A=\frac{4}{5}$ **B)** $\sin A=\frac{\sqrt{5}}{3}$ **C)** $\sin C=\frac{1}{3}$ **D)** $\sin C=\frac{1}{2\sqrt{2}}$

2. Base of a pyramid is rectangular, three of its vertices of the base are A (2, 2, −1), B(3, 1, 2) and C (1, 1, 1) and its vertex at the top is $P\left(4,\frac{-26}{3},\frac{-10}{3}\right)$. Fourth vertex of the base is D. Which of the following is **INCORRECT**?

A) If the coordinates of foot of normal drawn from P on the base of the pyramid are (ℓ,m,n) then $\ell+m-2n=2$.

B) If the coordinates of foot of normal drawn from P on the base of the pyramid are (ℓ,m,n) then $\ell>m>n$.

C) Volume of the pyramid is 20 cubic units.

D) Volume of the pyramid is 60 cubic units.

3. There are six students S_1,S_2,S_3,S_4,S_5 and S_6 in music class and for them there are six seats R_1,R_2,R_3,R_4,R_5 and R_6 arranged in a row, where initially the seat R_i is allotted to the student $S_i, i=1,2,3,4,5,6$. But, on the examination day, the six students are randomly allotted the six seats.Which of the following is **TRUE**?

Sri Chaitanya Educational Institutions
Infinity Learn BY SRI CHAITANYA
40 YEARS


For More Join: @IITJEE_Advanced

A) The number of arrangements on the examination day in which the student S_2 gets the previously allotted seat R_2 and none of the remaining students gets the seat previously allotted to him/her is 42.

B) The number of arrangements on the examination day in which the student S_2 gets the previously allotted seat R_2 and none of the remaining students gets the seat previously allotted to him/her is 53.

C) For i = 1, 2, 3, 4, 5. Let T_i denote the event that the students S_i and S_{i+1} do not sit adjacent to each other on the day of the examination then the number of arrangements of the event $T_1 \cap T_2 \cap T_3 \cap T_4 \cap T_5$ is 114.

D) For i = 1, 2, 3, 4, 5. Let T_i denote the event that the students S_i and S_{i+1} do not sit adjacent to each other on the day of the examination then the number of arrangements of the event $T_1 \cap T_2 \cap T_3 \cap T_4 \cap T_5$ is 90.

4. Consider $I = \int_1^e (\ln x) e^x dx$, then which of the following is **INCORRECT**?

A) $I > \frac{e^{e+1}}{2+2e}$ **B)** $I < \frac{e^{e+1}}{2+2e}$ **C)** $I < \left(e^{e+1} - e^e\right)$ **D)** $I < \frac{e^{e+1} - e^e}{2}$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

Sri Chaitanya Educational Institutions
Infinity Learn BY SRI CHAITANYA
40 YEARS


For More Join: @IITJEE_Advanced

5. If $S=\left\{x\in R:\sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right)-\sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)=\frac{\pi}{4}\right\}$, then which of the following is (are) **TRUE**?

A) The value of $\sum_{x\in S}\left[\sin\left(\left(x^2+x+5\right)\frac{\pi}{2}\right)-\cos\left(\pi\left(x^2+x+5\right)\right)\right]$ is equal to 2.

B) The value of $\sum_{x\in S}\left[\sin\left(\left(x^2+x+5\right)\frac{\pi}{2}\right)-\cos\left(\pi\left(x^2+x+5\right)\right)\right]$ is equal to 4.

C) The value of $\sum_{x\in S}\left[\cos\left(\left(x^2+x+4\right)\frac{\pi}{2}\right)-\sin\left(x^2+x+5\right)\pi\right]$ is equal to 4.

D) The value of $\sum_{x\in S}\left[\cos\left(\left(x^2+x+4\right)\frac{\pi}{2}\right)-\sin\left(\left(x^2+x+5\right)\pi\right)\right]$ is equal to 2.

6. The number of ordered pairs (a, b) of integers such that $\log_a b+6\log_b a=5,\ 2\le a\le 2026$ and $2\le b\le 2026$ is R, then R is not divisible by

A) 4 **B)** 5 **C)** 11 D) 13

7. Let A be a matrix of order $3\times 3, A=\left[a_{ij}\right]_{3\times 3}, 1\le i,j\le 3$ satisfying the condition

$A^{2025}+A=\begin{bmatrix}2&2&0\\0&2&2\\0&0&2\end{bmatrix}$, then which of the following statement(s) is (are) **CORRECT**?

(Where Tr(A) = Trace of matrix A, $|A|$ = determinant of matrix A)

A) $Tr\left(A^{2025}\right)=3$ **B)** $\left|A^{2026}\right|=1$

C) $a_{12}+a_{23}=\frac{1}{506}$ **D)** $a_{13}=\frac{-(2025)(1012)}{(2)(1013)^3}$

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and theon-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:+4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

8. If $f(x)$ be an invertible function $(x > 0)$ such that $3f(x) = 3x^4 + x^3 + 3x^2$ and $\int_3^{99} \frac{1}{\left(f^{-1}(x)\right)^2 + \left(f^{-1}(x)\right)^4} dx = \log_e n$, then the value of $(n-41)$ is equal to _____

9. Let C_1 and C_2 be two circles intersecting at points P and Q. The tangent line closer to Q touches C_1 and C_2 at M and N respectively. If PQ = 3, QN = 2, and MN = PN, The value of QM^2 is equal to _____

10. If ω is an imaginary cube root of unity and $\left(a + b\omega + c\omega^2\right)^{2026} = a + b\omega^2 + c\omega$ where a, b, c are unequal real numbers, then $a^2 + b^2 + c^2 - ab - bc - ca$ is equal to _____

11. A chord AB which is bisected at (1, 1) is drawn to the hyperbola $7x^2 + 8xy - y^2 - 4 = 0$ with centre C, which intersects the asymptotes in E and F. If the equation of circumcircle of ΔleCEF is $x^2 + y^2 - ax - by + c = 0$ then the value of $\frac{11a + 21b + 7c}{14}$ is equal to _____

12. Let f be continuous and the differentiable function g is defined as $g(x) = \int_0^x \left(t^2 \int_1^t f(u)\,du \right) dt$, where f(1) = 3, then the value of $g'(1) + g''(1)$ is

13. There are real numbers a, b, c and d such that –20 is a root of $x^3 + ax + b$ and –21 is a root of $x^3 + cx^2 + d$. These two polynomials share a complex root $m + i\sqrt{n}$, where m and n are positive integers and $i = \sqrt{-1}$, then the value of $\frac{n - 2m}{100}$ is equal to _____

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists :**List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

14. Consider the given data with frequency distribution.

x_i	2	7	9	5	3
f_i	4	8	3	k	3

Match each entry in Column-I with entries in Column –II.

Column–I		**Column–II**	
P)	If k = 7, then mean of the above data is	**1)**	1.75
Q)	If the mean of the above data is 5.5, then median is	**2)**	5.4
R)	If k = 2, then mean deviation about mean of the above data is	**3)**	2.25
S)	If the mean of the above data is 5.25, then mode of above data is	**4)**	5
		5)	7

A) P→2; Q→5; R→3; S→4 **B)** P→2; Q→3; R→5; S→4

C) P→4; Q→5; R→3; S→5 **D)** P→2; Q→4; R→1; S→5

15. Let $\vec{a},\vec{b},\vec{c},\vec{d}$ are position vectors of 4 points A, B, C, D such that $|\vec{a}|=1, |\vec{b}|=2, |\vec{c}|=3, |\vec{d}|=4$ and $(\vec{a}\times\vec{b}).(\vec{c}\times\vec{d})=24, \vec{a}.\vec{c}=\frac{3}{2}, \vec{b}.\vec{c}>0$, then area of $\Delta ABC=\frac{1}{4}\left(2\alpha+\beta\sqrt{3}\right)$ and area of $\Delta BCD=\frac{1}{2}\left(m-n\sqrt{3}\right)$, where $\alpha,\beta,m,n\in N$. Match List–I with List–II and select the **CORRECT** answer using the code given below the list.

List–I		**List–II**	
P)	α	1)	1
Q)	β	2)	3
R)	m	3)	4
S)	n	4)	9

A) P–1, Q–2, R–3, S–4 **B)** P–2, Q–1, R–3, S–4

C) P–2, Q–1, R–4, S–3 **D)** P–1, Q–2, R–4, S–3

Sri Chaitanya Educational Institutions
Infinity Learn BY SRI CHAITANYA
40 YEARS


For More Join: @IITJEE_Advanced

16. Consider $I_1=\int_0^1 \frac{e^{\tan^{-1}x}}{\sqrt{1+x^2}}dx, I_2=\int_0^1 \frac{xe^{\tan^{-1}x}}{\sqrt{1+x^2}}dx, I_3=\int_0^1 \frac{x^2e^{\tan^{-1}x}}{\sqrt{1+x^2}}dx$

List–I		List–II	
I)	I_1+I_2	P)	$\frac{3}{2}$
II)	$I_1+I_2-4I_3$	Q)	1
III)	$\frac{I_1+I_2+1}{I_1+I_2+2I_3}$	R)	$\sqrt{2}e^{\pi/4}-1$
IV)	$\frac{2(I_1+I_2)}{2I_3+1}$	S)	$\sqrt{2}e^{\pi/4}$
		T)	$\sqrt{2}e^{\pi/4}-3$

Then **CORRECT** option is

A) I–R, II–T, III–Q, IV–R **B)** I–S, II–T, III–P, IV–R

C) I–R, II–S, III–T, IV–P **D)** I–P, II–T, III–R, IV–S

17. Match Column-I with Column-II and select the correct answer using the code given below the list.

Column–I		Column–II	
A)	Let $\alpha,\beta,\gamma,\delta$ be the roots of the equation $x^4-x^3+2x^2+x+1=0$, then the value of $(\alpha^3+1)(\beta^3+1)(\gamma^3+1)(\delta^3+1)$ is	P)	0
B)	The value of $\frac{2^{180}}{45}\prod_{k=1}^{179}\sin k^\circ$ is	Q)	4
C)	If complex numbers z_1,z_2 and z_3 are the vertices of an isosceles right angled triangle, right angled at the vertex z_2, then $(z_1-z_2)^2+(z_3-z_2)^2$ equals	R)	8
D)	Let $Z=\frac{1}{2}(\sqrt{3}-i)$. If the least positive integer value of 'n' such that $\left(z^{101}+i^{109}\right)^{106}=z^n$ is 'k' then $\frac{2}{5}k$ is	S)	16

A) A–Q, B–R, C–P, D–S **B)** A–S, B–P, C–R, D–Q

C) A–S, B–R, C–P, D–Q **D)** A–Q, B–P, C–R, D–S

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

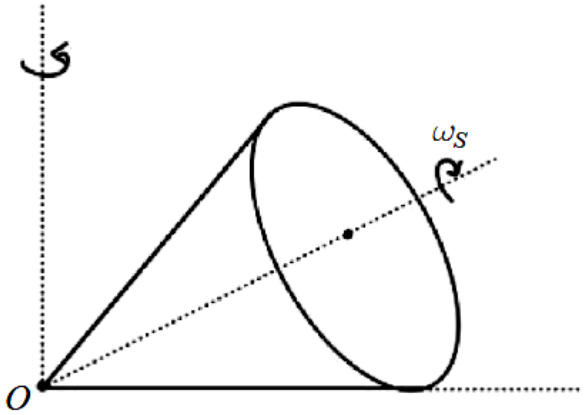
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

18. A point source emits sound equally in all directions in a non–absorbing homogeneous medium. Two points P and Q are at distances of 9 m and 25 m respectively from the source. The ratio of the amplitudes of wave at P and Q is

A) $\frac{3}{5}$ **B)** $\frac{5}{3}$ **C)** $\frac{9}{25}$ **D)** $\frac{25}{9}$

19. A uniform solid cone of mass m = 1 kg, radius R = 1m, height H = 1m is laid on a horizontal surface as shown in the figure, and set rolling without slipping on surface such that it spins about it's own axis with angular velocity $\omega_s = 1\frac{rad}{s}$ and process about a vertical axis passing through its apex O.



Magnitude of angular momentum (in SI units) of cone about O is

A) $\frac{5\sqrt{13}}{80}$ **B)** $\frac{7\sqrt{41}}{360}$ **C)** $\frac{3\sqrt{29}}{40}$ **D)** None of these

20. A capacitor is charged until its stored energy is 3J and then charging battery is removed. An another uncharged capacitor is now connected across it and it is found that charge distributes equally. The final value of total energy stored in the electric field is

A) 1.5 J **B)** 3 J **C)** 2.5 J **D)** 2 J

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

21. A plane progressive harmonic wave is given by the equation : $\phi = \phi_m \sin(2t - 3x + 4y + \pi/3)$, where x and y are in meters, and t is in seconds. Let $\hat{n}$ is the unit vector in the direction of wave propagation, and v is the speed of wave with respect to medium, then

A) $\hat{n} = -\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}; v = 0.4\,m/s$ **B)** $\hat{n} = \frac{4}{5}\hat{i} - \frac{3}{5}\hat{j}; v = \frac{2}{3}m/s$

C) $\hat{n} = \frac{3}{5}\hat{i} - \frac{4}{5}\hat{j}; v = 0.4\,m/s$ **D)** $\hat{n} = -\frac{4}{5}\hat{i} + \frac{3}{5}\hat{j}; v = 0.5\,m/s$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

22. Two particles of same mass m are connected by an ideal string of length ℓ. Consider system to be present in free space and string in just taut. Now an impulse is imparted to particle A in direction perpendicular to string. Due to this, instantaneously it starts moving with speed u. In subsequent motion.



A) Magnitude of tension in string remains constant.

B) In the frame of reference of particle A, the particle B performs periodic motion with period $T = \frac{2\pi\ell}{u}$.

C) In the frame of reference of particle A, the particle B performs periodic motion with period $T = \frac{\pi\ell}{u}$.

D) In a general interval of time $t = t$ to $t = t + T$, the particle B travels a distance equal to 4ℓ.

Sri Chaitanya Educational Institutions
Infinity Learn BY SRI CHAITANYA
40 YEARS


For More Join: @IITJEE_Advanced

23. There is a long straight wire carrying a STEADY current. An observer A who is stationary with respect to wire is standing at distance r from wire. Another observer B is moving along a path parallel to wire at distance r from it with non–relativistic velocity $V(<<c)$ in the direction of electric current in wire. Then choose the **CORRECT** statement(s)

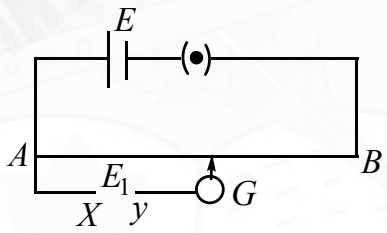
A) Electric current in wire in frames of A and B are equal.

B) Magnetic field at a point in frames of A and B are equal.

C) Electric field at a point in frame of A and B are equal.

D) Linear charge density of wire in frames of A and B are equal.

24. While doing an experiment with potentiometer (shown in fig.) it was found that the deflection is one sided and (i) the deflection decreases while moving the jockey from end A of the wire to the end B; (ii) the deflection increases while the jockey was moved from end A towards the end B.



Choose the correct option(s)

A) In case (i), E_1 will be greater than E.

B) In case (ii), E_1 will be greater than E.

C) In case (i) positive terminal of cell is connected to X and negative to Y.

D) In case (ii), negative terminal of cell is connected to X and positive to Y.

SECTION-III
(NON-NEGATIVE INTEGER)

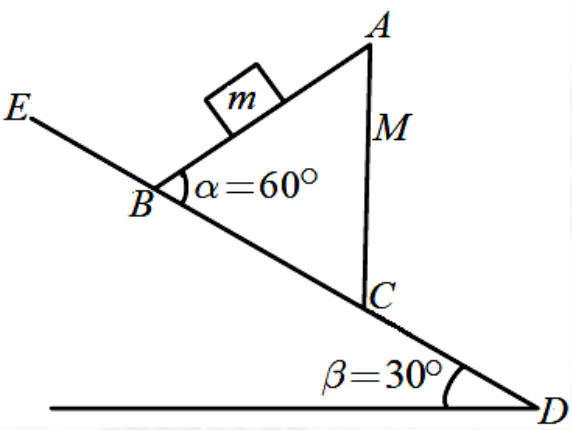
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and theon-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:+4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

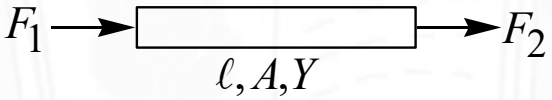
Sri Chaitanya Educational Institutions
Infinity Learn BY SRI CHAITANYA
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 SINGARAJU VENKAT KOUNDINYA
JEE Advanced 2023 RANK 1 VAVILALA CHIDVILAS REDDY
NEET 2023 RANK 1 BORA VARUN CHAKRAVARTHI
For More Join: @IITJEE_Advanced

25. A smooth wedge ABC of mass M can slide freely on a smooth fixed plane DE inclined at an angle β. On face AB of the wedge, there is a small block of mass m. What is acceleration of wedge ABC with respect to ground in m/s^2?

(M = 2 kg, m = 4 kg, $g = 10\,m/s^2$)



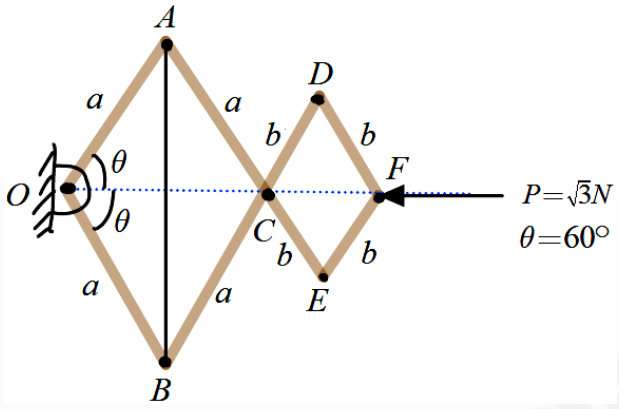
26. A thin uniform rod of length $\ell = 8\,m$, cross sectional area $A = 1\times10^{-4}\,m^2$ and Young's modulus $Y = 2\times10^{8}\,\frac{N}{m^2}$ is pushed by force $F_1 = 100\,N$ at left end and pulled by $F_2 = 200\,N$ at right end. Due to this pushing and pulling force, elongation of rod (in cm) is



27. A pantograph is a mechanical linkage connected in a manner based on parallelogram. The pantograph shown in the diagram is constructed of 3 pairs of identical light rods which are hinged at their ends and joints, forming two rhombii having ratio of their sides as a : b = 3 : 2. A horizontal force $P = \sqrt{3}N$ is applied at F. Find the tensile force (in N) developed in the string connected the two hinges A and B. (Assume all the hinges to be PIN joints).

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced



28. A photoelectric material having work–function ϕ_0 is illuminated with light of wavelength $\lambda\left(\lambda < \frac{hc}{\phi_0}\right)$. The fastest photoelectron has a de Broglie wavelength λ_d. A change in wavelength of incident light by $\Delta\lambda$ results in a change $\Delta\lambda_d$ in λ_d. Then the ratio $\frac{\Delta\lambda_d}{\Delta\lambda}$ is proportional to $\frac{\lambda_d^m}{\lambda^n}$, here m and n are co–prime numbers. Find value of m + n $\left(\Delta\lambda_d << \lambda_d \text{ and } \Delta\lambda << \lambda\right)$.

29. A fixed container of height 'H' with large cross–sectional area 'A' is completely filled with water. Two small orifices of cross sectional area 'a' each are made, one at the bottom and other on the vertical side of the container at a distance $\frac{H}{2}$ from the top of the container. The time taken by the water level to reach a height of $\frac{H}{2}$ from the bottom of the container is $t = \frac{2A}{na}\left(\sqrt{2}-1\right)\sqrt{\frac{H}{g}}$. Find 'n'.

30. Three particles, each of mass m, are situated at the vertices of an equilateral triangle of side length a. The only forces acting on the particles are their mutual gravitational forces. It is desired that each particle moves in a circle while maintaining the original mutual separation a. The time period of the circular motion is $2\pi\sqrt{\frac{a^3}{xGm}}$, then find the x value?

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists :**List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

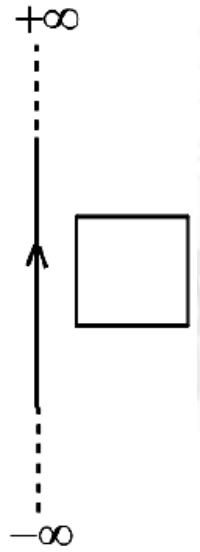
31. An ideal monoatomic gas undergoes different types of processes which are described in Column–I. Match the corresponding effects in Column–II. The symbols have usual meaning.

	Column–I		Column–II
A)	$PV^{-2} = \text{constant}$	**P)**	If volume increases then temperature will also increase.
B)	$PV^{2} = \text{constant}$	**Q)**	If volume increases then temperature decreases.
C)	$C = C_v + 2R$	**R)**	For expansion, heat will have to be supplied to the gas.
D)	$C = C_v - 2R$	**S)**	If temperature increases then work done by gas is positive.
		T)	If pressure increases the work done by gas is negative.

A) A–PS, B–QT, C–RST, D–QRT **B)** A–QT, B–PRS, C–PQRS, D–QT

C) A–PRS, B–QT, C–PRST, D–QRT **D)** A–PRS, B–PS, C–RST, D–PS

32. A conducting square loop is placed near a long straight current carrying wire as shown. Then match the two columns.



Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS
THE PERFECT HAT-TRICK WITH ALL– INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

	Column–I		Column–II
A)	If current is increased	P)	Induced current in loop is clockwise.
B)	If current is decreased	Q)	Induced current in loop is anticlockwise.
C)	If loop is moved away from the wire	R)	Wire will attract the loop.
D)	If loop is moved toward the wire	S)	Wire will repel the loop

A) A–QS, B–PR, C–QS, D–PR **B)** A–QS, B–PR, C–PR, D–QS

C) A–PR, B–QS, C–QS, D–PR **D)** A–PR, B–QS, C–PR, D–QS

33. Match Column–I of the nuclear processes with Column–II containing parent nucleus and one of the end products of each process. Then select the correct answer using the codes given below the columns:

	Column–I		Column–II
A)	Alpha decay	P)	${}^{15}_{8}O \rightarrow {}^{15}_{7}N + ...$
B)	Positron emission	Q)	${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + ...$
C)	Nuclear fission	R)	${}^{185}_{83}Bi \rightarrow {}^{184}_{82}Pb + ...$
D)	Proton emission	S)	${}^{239}_{94}Pu \rightarrow {}^{140}_{57}La + ...$

A) A–Q, B–P, C–S, D–R **B)** A–P, B–Q, C–S, D–R

C) A–R, B–P, C–S, D–P **D)** A–S, B–R, C–Q, D–P

34. For an object moving along the principal axis of an optical system, match the conditions given in Column–I to the optical system given in Column–II.

	Column–I		Column–II
A)	Object and image move in same direction.	P)	Converging mirror.
B)	Object and image move in opposite direction.	Q)	Diverging mirror.
C)	If a real object moves towards the optical centre/pole, its image will always move towards optical centre/pole.	R)	Converging lens.
D)	If a virtual object moves towards the optical centre/pole its image will always move towards optical centre/pole.	S)	Diverging lens.

A) A–RS, B–PQ, C–QS, D–PR **B)** A–RS, B–PQ, C–PQ, D–PR

C) A–PQ, B–RS, C–QS, D–PR **D)** A–RS, B–PR, C–QS, D–PQ

Sri Chaitanya Educational Institutions
Infinity Learn BY SRI CHAITANYA
40 YEARS


For More Join: @IITJEE_Advanced

CHEMISTRY **Max Marks: 60**

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

35. A pure amino acid X is analyzed in two experiments.

Experiment I — Combustion

A 0.150 g sample of amino acid X is completely burnt, producing, 0.282 g of CO_2 and 0.127 g of H_2O .Nitrogen is 12.00% by mass in X. Assume X contains one $-NH_2$, one –COOH, and no ionisable side-chain.

Experiment II — Acid–Base Properties of Zwitter ionic form

Half-equivalence pH with HCl = 2.20.

Half-equivalence pH with NaOH = 9.80.

Which option correctly identifies the amino acid and the pH at which its net charge is zero?

A) Alanine, pI = 6 **B)** Glycine, pI = 7 **C)** Valine, pI = 6 **D)** Leucine, pI = 4

36. Consider the following radial wavefunctions (ψ_r) corresponding to different hydrogen–like orbitals:

i) $\psi_r = 2\left(\frac{1}{a_0}\right)^{3/2} e^{-r/a_0}$

ii) $\psi_r = \frac{1}{9\sqrt{3}}\left(\frac{1}{a_0}\right)^{3/2}\left(6-6\sigma+\sigma^2\right)e^{-\sigma/2}, \sigma = \frac{2r}{na_0}$

iii) $\psi_r = \frac{1}{2\sqrt{2}}\left(\frac{1}{a_0}\right)^{3/2}\left(2-\frac{2r}{na_0}\right)e^{-\frac{r}{na_0}}$

iv) $\psi_r = \frac{1}{9\sqrt{6}}\left(\frac{1}{a_0}\right)^{3/2}(4-\sigma)\sigma e^{-\sigma/2}, \sigma = \frac{2r}{na_0}$

Which of the following is incorrect?

Sri Chaitanya Educational Institutions
Infinity Learn BY SRI CHAITANYA
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA sri chaitanya 6th-12th Class 300 RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY sri chaitanya 6th-12th Class 341 360 RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI sri chaitanya 6th-12th Class 720 RANK 1
For More Join: @IITJEE_Advanced

A) For orbital (iv), the graph of (ψ_r) vs r intersects the x–axis at two distinct, non–zero values of r.

B) For orbital (i), the radial probability density is maximum at $r = 0$.

C) For orbital (ii), there is no angular node.

D) For orbital (iii), the distance of the radial node from the orign is na_0.

37. An ionic crystalline solid MX_2 has a cubic unit cell. Which one of the following arrangements of ions is consistent with the stoichiometry of the compound?

A) Tetravalent metal ion (M^{+4}) forms face centered cubic lattice and all tetrahedral voids are occupied by divalent anions

B) Divalent metal ion (M^{+2}) forms face centered cubic lattice and half of octahedral voids are occupied by mono valent anions

C) Anion forms body centered cubic lattice and metal ion occupies all distorted octahedral voids

D) Tetravalent metal ion (M^{+4}) forms body centered cubic lattice and all distorted octahedral voids are occupied by anions

38. Which of the following sets of compounds all have same bond angles about central atom?

A) $CO_2, CS_2, CH_2O, CN_2^{2-}$ **B)** $C\ell O_3^-, BrO_3^-, XeO_3, SO_3^{2-}$

C) $BF_4^-, NH_4^+, CH_4, SiH_4$ **D)** $SO_2C\ell_2, POC\ell_3, C\ell O_4, XeO_4$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

Sri Chaitanya Educational Institutions
Infinity Learn BY SRI CHAITANYA
40 YEARS


For More Join: @IITJEE_Advanced

39. Consider the complex $\left[Co(en)_2 C\ell_2\right]^{+1}$ choose the correct statement(s):

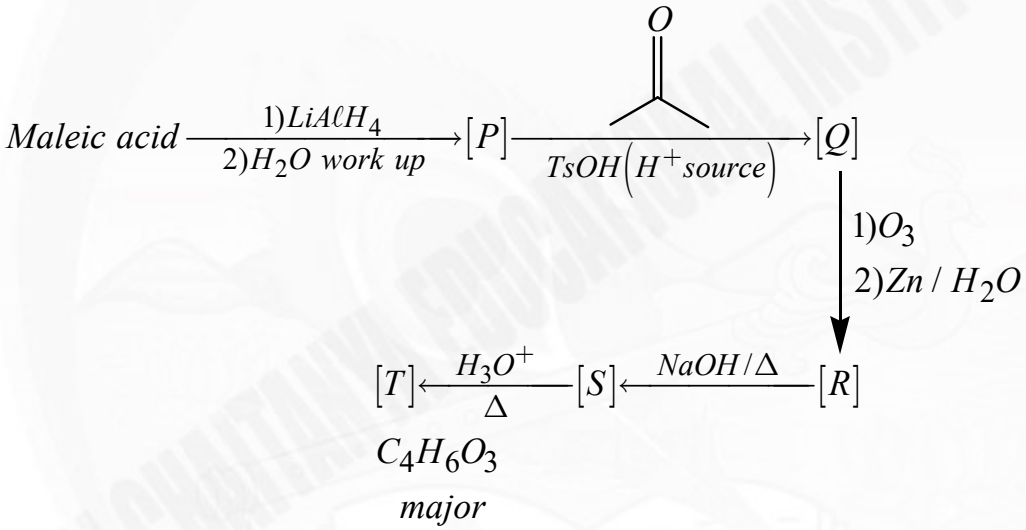
A) All stereoisomers are repelled by external magnetic field.

B) Trans isomer and cis isomer have different colours.

C) Crystal field stabilization energy is $\left(-2.4\Delta_o + 3P\right)$ where P = Pairing Energy and Δ_o = octahedral splitting constant

D) Cis isomer is optically resolvable while trans isomer is not.

40. Choose the CORRECT option(s) regarding the given scheme.



A) **[T]** is chiral

B) **[T]** gives positive Tollen's test

C) **[T]** gives positive Fehling's test

D) **[T]** upon reaction with NH_2OH (excess) gives more than two stereoisomers

41. Identify the option(s) that is(are) NOT correct regarding adsorption:

A) Adsorption causes an increase in the unbalanced (residual) attractive forces acting on the molecules present on the surface.

B) For a fixed mass of an adsorbent, the amount of gas adsorbed per unit mass remains unchanged even if the surface area of that mass is altered.

C) At adsorption equilibrium, change in entropy of universe is zero.

D) The adsorption of a gas on a solid surface is primarily driven by an increase in entropy.

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

SECTION-III
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

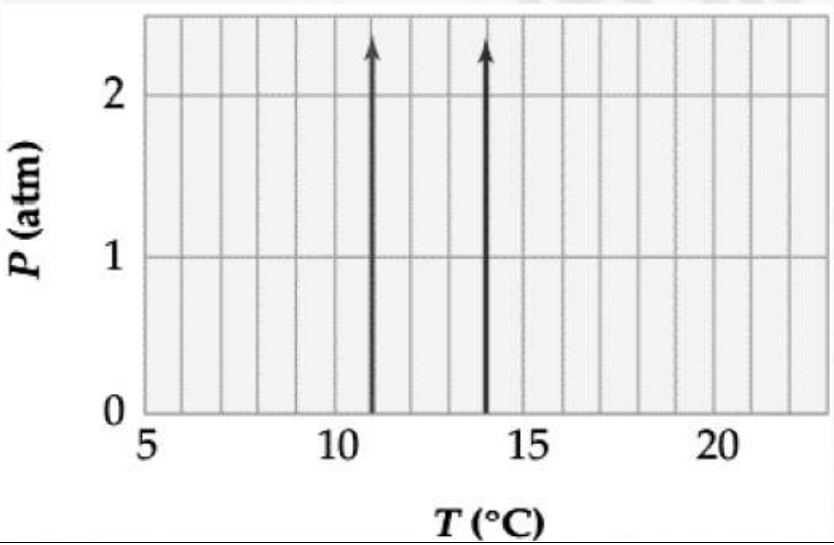
42. The K_{sp} of $Ag_2C_2O_4$ at $25^\circ C$ is $1.25\times10^{-11} mol^3L^{-3}$. A solution of $K_2C_2O_4$ containing 1.485 mole in 500 mL water is shaken at $25^\circ C$ with excess Ag_2CO_3 until the equilibrium is reached.

$$Ag_2CO_{3(s)} + K_2C_2O_{4(aq)} \rightleftharpoons Ag_2C_2O_{4(s)} + K_2CO_{3(aq)}$$

At equilibrium the solution contains 0.36 mole of K_2CO_3. Assuming degree of dissociation of $K_2C_2O_4$ and K_2CO_3 are same, calculate K_{sp} of Ag_2CO_3 in mol^3L^{-3}.

If $K_{sp} = x\times10^{-y}$ in scientific notation then what is the value of $\frac{y}{x}$?

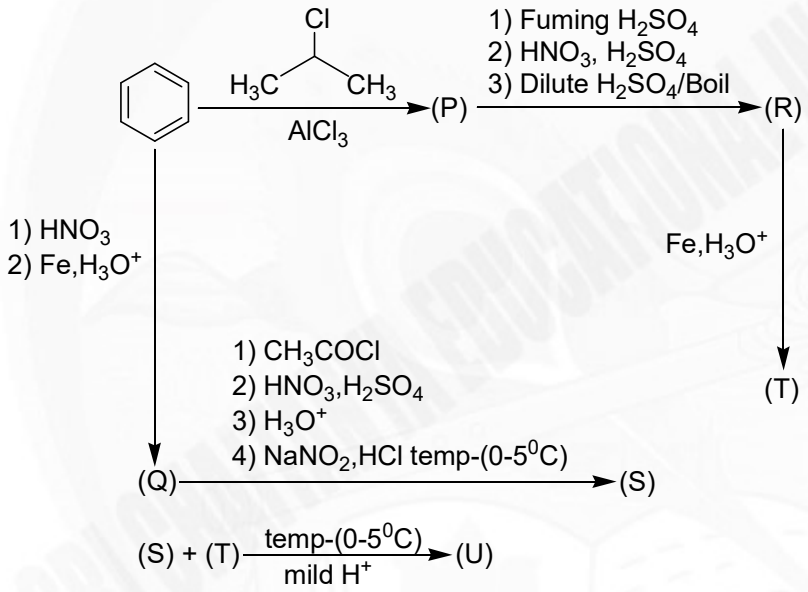
43. The following phase diagram shows a very small part of solid-liquid phase transition boundaries for two solutions of equal concentrations. Van't Hoff factors for substance A and B are 1 and 3 respectively. If melting point of pure liquid in centigrade scale is x, the value of 2x is _____



Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

44. When the compound (X) is treated with sodium iodate and hydrochloric acid a brown coloured solution (or violet coloured substance) (Z) is produced. The colour of Z disappears when treated with the compound (A) producing (X) and (X'). (A) is produced by reaction between sulphur and sodium sulphite. How many maximum number of bonds between identical atoms having equal bond length in the compound (X') is possible?

45.



Total number of Nitrogen atoms present in compound (U) is_____

46. The number of compounds which on heating gives dioxygen as one of the products is_____

$HgO, MnO_2, Pb_3O_4, KMnO_4, K_2Cr_2O_7, Ag_2CO_3, KClO_3, LiNO_3$

47. The enthalpies of neutralization of a weak acid HX and a weak acid HY by NaOH are −6900 cal/equivalent and −2900 cal/equivalent, respectively. When one equivalent of NaOH is added to a solution containing one equivalent of HX and one equivalent of HY, the enthalpy change was −3900 cal. If the base is distributed between HX and HY in the ratio 1: x, the value of 'x' is _____

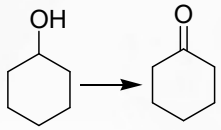
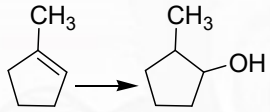
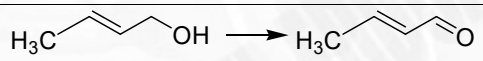
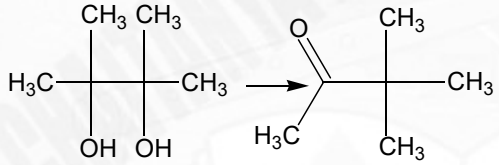
Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases..

48. Match the conversion in Column–I with appropriate reagent(s) in Column–II.

Column–I		Column–II	
A)		P)	$Conc.H_2SO_4$
B)		Q)	$Cu\,/\,573K$
C)		R)	i) $B_2H_6 - THF$ ii) $H_2O_2\,/\,OH^-$
D)		S)	PCC

A) A – RS, B – P, C – QS, D – QS **B)** A – QS, B – P, C – QS, D – Q
C) A – QS, B – R, C – QS, D – P **D)** A – PQ, B – R, C – QS, D – Q

49. Match the compound given in column-I with the property given in column-II.

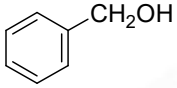
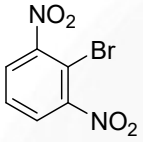
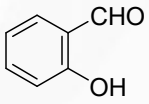
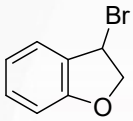
Column–I		Column–II	
A)	Graphite	P)	sp^2 hybridisation
B)	Boron nitride (h-BN)	Q)	Layered lattice structure
C)	Fullerene	R)	Bond order 1.33
D)	Boric acid	S)	Solid lubricant
		T)	Electrical conductor

A) A–PST, B–PQRS, C–P, D–QST
B) A–PQST, B–PQRS, C–PTS, D–PQS
C) A–PQRST, B–PQRST, C–PRT, D–QRT
D) A–PQRST, B–PQRS, C–P, D–PQ

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

50. Match the following List–I with the List–II.

Column–I		Column–II	
A)		P)	Nucleophilic substitution
B)		Q)	Nucleophilic addition
C)		R)	Esterification with $(CH_3CO)_2O$
D)		S)	Dehydrogenation on heating with Cu at $300^\circ C$
		T	Dehydrogenation

A) A–P,R,S, B–P, C–Q,R, D–P,T **B)** A–P,R, B–P,S, C–Q,R, D–Q,T

C) A–P, S, B–P, C–Q, D–P,R **D)** A–P,Q,S, B–P,R, C–Q,R, D–P,Q

51. Match the extraction methods (List–I) with the metals generally obtained by those methods (List–II). Choose the correct code.

Column–I		Column–II	
A)	Carbon reduction	P)	Silver
B)	Decomposition of volatile metal iodide	Q)	Lead
C)	Self–reduction from sulphide ore	R)	Boron
D)	Coordinate covalent compound formation followed by metal displacement	S)	Copper

A) A–Q, B–R, C–Q,S, D–P **B)** A–S, B–P, C–Q, D–R

C) A–R, B–Q, C–P, D–S D) A–P, B–S, C–R, D–Q

Sri Chaitanya Educational Institutions
Infinity Learn BY SRI CHAITANYA
40 YEARS


For More Join: @IITJEE_Advanced

Sri Chaitanya Educational Institutions | Infinity Learn

Sri Chaitanya Techno School
The right mentor for IIT (JEE), NEET, Olympiad & all Other Competitive exams

40 Years of Excellence 1986-2025

JEE MAIN 2025

31 STUDENTS BELOW 100 AIR



VANGALA AJAY REDDY
APP. NO. 250310255592
CLASSROOM STUDENT FROM GRADE IX - XII



DEVDUTTA MAJHI
APP. NO. 250310016185
DLP/AITS STUDENT



TOSHNIWAL SHIVEN
APP. NO. 250310391420
DLP/AITS STUDENT



SAKSHAM JINDAL
APP. NO. 250310236696
DLP/AITS STUDENT

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 10 ALL INDIA CATEGORY RANKS COUNT	BELOW 100 ALL INDIA CATEGORY RANKS COUNT	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT
31	95	10	98	579

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2025 22,094

*DLP/AITS

JEE 2025 STARS SHINE BRIGHT

Sri Chaitanya Tops JEE ADVANCED

ALL INDIA OPEN CATEGORY RANKS



RUTVIK SAI
H.T.No. 256055278 (OBC-NCL)



MAJID MUJAHID HUSAIN
H.T.No. 251134112*



UJJWAL KESARI
H.T.No. 252016104*



AKSHAT KUMAR CHAURASIA
H.T.No. 254065055*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	NUMBER OF QUALIFIED RANKS
29	113	205	745	4,212

*DLP/AITS

www.sichaitanya.net 040-66 06 06 06

For More Join: @IITJEE_Advanced